

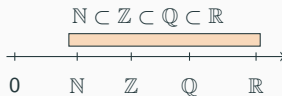
Math Bootcamp

Day 1 — Foundations (Sections 1.1–1.9)

Math Camp 2026

1.1 Number systems — names and definitions

Symbol	English name	What it is
$\mathbb{N}$	natural numbers	Non-negative integers for counting
$\mathbb{Z}$	integers	$\dots, -2, -1, 0, 1, 2, \dots$
$\mathbb{Q}$	rational numbers	Fractions p/q ($q \neq 0$)
$\mathbb{R}$	real numbers	All points on the number line
$\mathbb{R}^n$	n -dimensional real space	Lists / vectors of length n



1.1 Symbols and vectors — names and definitions

Common symbols	$\in$	“is an element of”	$\forall$	“for all”
	$\exists$	“there exists”	$\Rightarrow$	“implies”
			$\Leftrightarrow$	“if and only if” (iff)

Scalar vs. vector

- **Scalar:** one number, e.g. income $w \in \mathbb{R}$
- **Vector:** a list of numbers $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$
- **Dot product:** $\mathbf{p} \cdot \mathbf{x} = \sum_{i=1}^n p_i x_i$

1.1 — Which set? Together

Smallest set that fits?

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2. Money you spend at a vending machine (e.g. HK\$8.50)

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→ $\mathbb{Q}$

e.g. $8.50 = 17/2$ dollars

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3. Order as a vector: (HK\$ spent on drink, # snacks) = (8.50, 2)

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e.g. $8.50 = 17/2$ dollars

3. Order as a vector: (HK\$ spent on drink, # snacks) = (8.50, 2)

→ $(8.50, 2) \in \mathbb{Q} \times \mathbb{N}$

1.2 Logic — names and definitions

English	Symbol	Meaning
P only if Q	$P \Rightarrow Q$	Q is necessary for P (without Q , no P)
P if Q	$Q \Rightarrow P$	Q is sufficient for P (Q alone is enough)
P if and only if Q	$P \Leftrightarrow Q$	Q is necessary and sufficient
Contrapositive	$\neg Q \Rightarrow \neg P$	same truth as $P \Rightarrow Q$: if Q fails, P must fail

1.1 — IKEA run Together

First week in HK: you go to IKEA and spot three plush toys.



Otter HK\$179.9



Hedgehog HK\$79.9



BLÅHAJ HK\$179.9

You have **HK\$500** pocket money ($B = 500$).

Basket $\mathbf{x} \in \mathbb{N}^3$: # otters, # hedgehogs, # sharks. E.g. $(1, 1, 0)$ = one otter, one hedgehog, no shark. 6/46
Price vector $\mathbf{p} = (179.9, 79.9, 179.9) \in \mathbb{Q}^3$. Budget set: $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$.

1.2 — Plush symbols practice Together

$\mathbf{p} = (179.9, 79.9, 179.9)$, $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$. True or false?

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5. $[\Rightarrow]$ $(1, 1, 1) \in \mathcal{B}$ **only if** $\mathbf{p} \cdot (1, 1, 1) \leq 500$

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6. $[\Rightarrow]$ $(0, 0, 3) \in \mathcal{B}$ **if** $(1, 1, 1) \in \mathcal{B}$

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3. $[\forall]$ $\forall \mathbf{x} \in \mathcal{B}, \mathbf{x} \neq (0, 1, 0)$ **False** ($(0, 1, 0) \in \mathcal{B}$)
4. $[\exists]$ $\exists \mathbf{x} \in \mathcal{B}, \mathbf{x} = (3, 0, 0)$ **False** ($539.7 > 500$)
5. $[\Rightarrow]$ $(1, 1, 1) \in \mathcal{B}$ only if $\mathbf{p} \cdot (1, 1, 1) \leq 500$ **True** ($439.7 \leq 500$)

also True: $(1, 1, 1) \in \mathcal{B}$ if $\mathbf{p} \cdot (1, 1, 1) \leq 500$

6. $[\Rightarrow]$ $(0, 0, 3) \in \mathcal{B}$ if $(1, 1, 1) \in \mathcal{B}$ **False** ($(1, 1, 1) \in \mathcal{B}$ but $539.7 > 500$)

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6. $[\Rightarrow]$ $(0, 0, 3) \in \mathcal{B}$ **if** $(1, 1, 1) \in \mathcal{B}$ **False** ($(1, 1, 1) \in \mathcal{B}$ but $539.7 > 500$)
7. [contrapositive] **If** $\mathbf{p} \cdot (1, 1, 1) + \mathbf{p} \cdot (0, 1, 0) > 500$, **then** $(1, 1, 1) \notin \mathcal{B}$ **or** $(0, 1, 0) \notin \mathcal{B}$

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2. $[\subset] \mathbb{N}^3 \subset \mathcal{B}$ **False** (e.g. $(3, 0, 0) \notin \mathcal{B}$)
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also True: $(1, 1, 1) \in \mathcal{B}$ **if** $\mathbf{p} \cdot (1, 1, 1) \leq 500$

6. $[\Rightarrow] (0, 0, 3) \in \mathcal{B}$ **if** $(1, 1, 1) \in \mathcal{B}$ **False** ($(1, 1, 1) \in \mathcal{B}$ but $539.7 > 500$)
7. [contrapositive] If $\mathbf{p} \cdot (1, 1, 1) + \mathbf{p} \cdot (0, 1, 0) > 500$, then $(1, 1, 1) \notin \mathcal{B}$ **or** $(0, 1, 0) \notin \mathcal{B}$ **False** ($519.6 > 500$, yet each basket alone is in $\mathcal{B}$)

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also True: $(1, 1, 1) \in \mathcal{B}$ if $\mathbf{p} \cdot (1, 1, 1) \leq 500$

6. $[\Rightarrow] (0, 0, 3) \in \mathcal{B}$ if $(1, 1, 1) \in \mathcal{B}$ **False** ($(1, 1, 1) \in \mathcal{B}$ but $539.7 > 500$)
7. [contrapositive] If $\mathbf{p} \cdot (1, 1, 1) + \mathbf{p} \cdot (0, 1, 0) > 500$, then $(1, 1, 1) \notin \mathcal{B}$ or $(0, 1, 0) \notin \mathcal{B}$ **False** ($519.6 > 500$, yet each basket alone is in $\mathcal{B}$)

each affordable $\nRightarrow$ jointly affordable

1.3 Sets and operations — names and definitions

Symbol	Name	Meaning
$\in, \notin$	element / not in	x belongs (or not) to A
$\subset, =$	subset, equality	$A \subset B$; $A = B$ iff mutual $\subset$
$\emptyset, \{ \}, \{x \in S : \dots\}$	sets	empty, roster, set-builder
$A \cup B, A \cap B$	union, intersection	in A or B ; in both
$A^c, A \setminus B$	complement, difference	not in A ; in A not B
$A \times B$	Cartesian product	pairs (a, b) , $a \in A$, $b \in B$

In economics: budget set $\mathcal{B} \subset \mathbb{N}^3$; $(8.50, 2) \in \mathbb{Q} \times \mathbb{N}$.

1.3 — Set operations (IKEA) Together

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$, $\mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

Describe the set.

1.3 — Set operations (IKEA) Together

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Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$

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Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$ affordable baskets with ≥ 1 hedgehog (e.g. $(0, 1, 0)$)

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2. $\mathcal{B} \cup \mathcal{H}$

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Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$ affordable baskets with ≥ 1 hedgehog (e.g. $(0, 1, 0)$)
2. $\mathcal{B} \cup \mathcal{H}$ affordable baskets or baskets with ≥ 1 hedgehog
3. $\mathcal{B} \setminus \mathcal{H}$

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4. $(179.9, 79.9) \in \mathbb{Q} \times \mathbb{Q}$ but $(179.9, 79.9) \notin \mathcal{B}$

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4. $(179.9, 79.9) \in \mathbb{Q} \times \mathbb{Q}$ but $(179.9, 79.9) \notin \mathcal{B}$ True Cartesian product $\neq$ budget set — pairs $\neq$ baskets

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5. $(1, 0, 0) \in \mathcal{B}$

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3. $\mathcal{B} \setminus \mathcal{H}$ affordable baskets with no hedgehog (e.g. $(1, 0, 0)$)
4. $(179.9, 79.9) \in \mathbb{Q} \times \mathbb{Q}$ but $(179.9, 79.9) \notin \mathcal{B}$ **True** Cartesian product $\neq$ budget set — pairs $\neq$ baskets
5. $(1, 0, 0) \subset \mathcal{B}$ **False** $(1, 0, 0)$ is a basket: use $(1, 0, 0) \in \mathcal{B}$ (True); but $\{(1, 0, 0)\} \subset \mathcal{B}$ is True

1.4 Functions — names and definitions

Term	Notation	Meaning
Function	$f: A \rightarrow B$	each $x \in A$ maps to exactly one $f(x) \in B$
Domain	A	allowed inputs
Codomain	B	target set — every output must lie in B
Range	$f(A)$	outputs hit: $\{f(x) : x \in A\} \subseteq B$
Injective (one-to-one)		$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$
Surjective (onto)		every $y \in B$ equals some $f(x)$
Operations	$(f \pm g)(x)$, $(cf)(x)$, $(fg)(x)$	$(f \pm g)(x) = f(x) \pm g(x)$; $(cf)(x) = c f(x)$; $(fg)(x) = f(x)g(x)$
Composition	$(f \circ g)(x)$	apply g first: $(f \circ g)(x) = f(g(x))$

1.4 — Ages Together

Define $f: D \rightarrow C$ by: $f(x)$ = your girlfriend / boyfriend's age when **your age** is x .

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typical ages in a master's program

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6. Is f injective (one-to-one)? **Probably not** (e.g. $f(22) = f(30) = 22$)
7. If $f(25) = \{22, 80\}$, is f a function? **No — a correspondence** (a function gives **exactly one** output per input; it's immoral to have a correspondence ☺)

1.5 — Reservation price (example first) Together

BLÅHAJ sticker price: **HK\$179.9**. You buy **only on sale**: when $p < 179.9$.

Prices where you still buy: $P = \{p \in \mathbb{R}_+ : p < 179.9\}$.

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Reservation price (WTP)? $\sup P = 179.9$ least upper bound — the choke price)

Is $179.9 \in P$? **No** at sticker price you walk away $\Rightarrow$ no max P)

1.5 Supremum and infimum — names and definitions

Term	Notation	Meaning
Maximum	$\max S$	largest element in S
Minimum	$\min S$	smallest element in S
Supremum	$\sup S$	least upper bound of S
Infimum	$\inf S$	greatest lower bound of S

Example (least upper bound): $S = [0, 1)$. Upper bounds include $1, 2, 3, \dots$; $\sup S = 1$ — the **smallest** upper bound ($1 \notin S$).

Relations: if $\max S$ exists, $\max S = \sup S$; if $\min S$ exists, $\min S = \inf S$.

Always: $\inf S \leq x \leq \sup S$ for all $x \in S$.

1.6 Taylor, logs, and exponentials

Taylor expansion (around a):

$$f(a+h) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} h^k = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \frac{f^{(3)}(a)}{3!}h^3 + \dots$$

Logs

$$\ln(ab) = \ln a + \ln b$$

$$\ln(x^\alpha) = \alpha \ln x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

Exponentials

$$e^{a+b} = e^a e^b$$

$$e^{\alpha x} = (e^x)^\alpha$$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots$$

Quick check: $\ln(x_1^\alpha x_2^\beta) = \alpha \ln x_1 + \beta \ln x_2$; if $u = x_1^\alpha x_2^{1-\alpha}$, then $\ln u = \alpha \ln x_1 + (1-\alpha) \ln x_2$.

1.6 Why logs?

- **Taylor:** $f(a + h) \approx f(a) + f'(a)h$.
- **Apply to logs:** $\ln(1 + x) \approx x$ for small x .
- **Percentage changes:** $\ln(x + \Delta x) - \ln x = \ln\left(1 + \frac{\Delta x}{x}\right) \approx \frac{\Delta x}{x}$, so $\Delta \ln x$ tracks % change in x .
- **Example:** $\frac{\Delta x}{x} = 5\% \Rightarrow \Delta \ln x \approx 0.05$.
- **Log-log regression:** $\ln y = \alpha + \beta \ln x$.
- $1\% \uparrow$ in $x \Rightarrow \beta\%$ change in y .
- **Elasticity:** $\varepsilon_{y,x} = \frac{\partial y}{\partial x} \frac{x}{y} = \frac{\Delta y/y}{\Delta x/x} = \frac{\partial \ln y}{\partial \ln x}$; in $\ln y = \alpha + \beta \ln x$, $\varepsilon_{y,x} = \beta$.

1.6 Cobb–Douglas — definition

Utility / production:

$$U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}, \quad Y(K, L) = AK^\alpha L^{1-\alpha}, \quad \alpha \in (0, 1).$$

In logs (linear): $\ln U = \alpha \ln x_1 + (1 - \alpha) \ln x_2$; $\ln Y = \ln A + \alpha \ln K + (1 - \alpha) \ln L$.

Constant elasticity: $\varepsilon_{U, x_1} = \frac{\partial \ln U}{\partial \ln x_1} = \alpha$, $\varepsilon_{U, x_2} = 1 - \alpha$; $\varepsilon_{Y, K} = \alpha$, $\varepsilon_{Y, L} = 1 - \alpha$. Same numbers at every (x_1, x_2) or (K, L) .

1.6 Why Cobb–Douglas? — first-order approximation

Any smooth utility $u(x_1, x_2) > 0$: define $f(\ln x_1, \ln x_2) = \ln u(x_1, x_2)$.

First-order Taylor around $(\ln \bar{x}_1, \ln \bar{x}_2)$:

$$\ln u(x_1, x_2) \approx f(\ln \bar{x}_1, \ln \bar{x}_2) + \left. \frac{\partial f}{\partial \ln x_1} \right|_{\bar{x}} (\ln x_1 - \ln \bar{x}_1) + \left. \frac{\partial f}{\partial \ln x_2} \right|_{\bar{x}} (\ln x_2 - \ln \bar{x}_2).$$

Let $\alpha_i = \left. \frac{\partial \ln u}{\partial \ln x_i} \right|_{\bar{x}}$ and absorb constants into c :

$$\ln u(x_1, x_2) \approx c + \alpha_1 \ln x_1 + \alpha_2 \ln x_2 \quad \Leftrightarrow \quad u(x_1, x_2) \approx e^c x_1^{\alpha_1} x_2^{\alpha_2}.$$

Near $\bar{x}$ only; Cobb–Douglas is exact when $\ln u$ is globally affine in logs.

1.6 Cobb–Douglas — homogeneity and CRS

Homogeneous of degree k : $f(tx) = t^k f(x)$ for $t > 0$.

Utility $U = x_1^\alpha x_2^{1-\alpha}$: $U(tx_1, tx_2) = t^\alpha t^{1-\alpha} U(x) = t U(x) \Rightarrow$ degree 1.

Production $Y = AK^\alpha L^{1-\alpha}$: $Y(tK, tL) = t^\alpha t^{1-\alpha} Y(K, L) = t Y(K, L)$.

CRS (constant returns to scale): double all inputs $\Rightarrow$ double output (homogeneous of degree 1).

1.6 Cobb–Douglas — constant expenditure shares

$$\text{Max } U = x_1^\alpha x_2^{1-\alpha} \text{ s.t. } p_1 x_1 + p_2 x_2 = w.$$

$$\text{FOC (marginal utility per dollar equal): } \frac{\partial U / \partial x_1}{\partial U / \partial x_2} = \frac{p_1}{p_2} \Rightarrow \frac{\alpha / x_1}{(1 - \alpha) / x_2} = \frac{p_1}{p_2} \Rightarrow \frac{p_1 x_1}{p_2 x_2} = \frac{\alpha}{1 - \alpha}.$$

$$\text{Budget binds: } p_1 x_1 + p_2 x_2 = w. \text{ Substitute } p_1 x_1 = \frac{\alpha}{1 - \alpha} p_2 x_2:$$

$$w = p_2 x_2 \left(\frac{\alpha}{1 - \alpha} + 1 \right) = \frac{p_2 x_2}{1 - \alpha} \Rightarrow p_2 x_2 = (1 - \alpha) w, \quad p_1 x_1 = \alpha w.$$

$$\text{Expenditure shares: } \frac{p_1 x_1}{w} = \alpha, \quad \frac{p_2 x_2}{w} = 1 - \alpha \text{ (constant).}$$

1.6 Why Cobb–Douglas? — summary

Why economists use Cobb–Douglas:

- **Logs = % changes:** elasticities are $\partial \ln y / \partial \ln x$.
- **Local approximation:** any smooth $u > 0$ satisfies $\ln u \approx c + \alpha_1 \ln x_1 + \alpha_2 \ln x_2$ near $\bar{\mathbf{x}}$.
- **Constant elasticity / shares:** $\varepsilon_{u, x_i} = \alpha_i$; CD demands give fixed budget shares.
- **Homogeneity (degree 1) / CRS:** $f(t\mathbf{x}) = tf(\mathbf{x})$; double inputs $\Rightarrow$ double output.
- **Easy algebra:** linear in logs, powers in levels; closed-form demands and FOCs.
- **Global use = modeling choice:** tractability and homogeneity, not a theorem for all preferences.

1.7 Single-variable derivative — formal definition

Scope: one independent variable. Partial derivatives ($\partial f / \partial x_i$) come later.

Definition. For $f: \mathbb{R} \rightarrow \mathbb{R}$ and $a \in \mathbb{R}$, the **derivative at a** is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

provided the limit exists. Same number as $\left. \frac{df}{dx} \right|_{x=a}$.

Meaning: instantaneous rate of change; slope of the tangent at $(a, f(a))$.

In economics: $MC = C'(q)$, $MU = U'(x)$, $MP_K = Y'(K)$.

1.7 Single-variable rules — summary

Single-variable calculus: all rules below are for $f(x)$, $g(x)$, etc. Built from

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad (\text{derivations in appendix):}$$

$$(c)' = 0$$

constant

$$(f \pm g)' = f' \pm g'$$

sum / difference

$$(x^n)' = nx^{n-1}$$

power rule

$$(fg)' = f'g + fg'$$

product rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

quotient rule

$$(e^x)' = e^x, (\ln x)' = \frac{1}{x}$$

exp / log

$$(a^x)' = a^x \ln a$$

general exponential ($a > 0$)

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$$

inverse function

Chain rule (composition): if $y = f(u)$, $u = g(x)$, then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$.

1.7 Single-variable chain rule

Chain rule: if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (f' \circ g)(x) g'(x).$$

Example: $y = (3x + 1)^5$. Let $u = 3x + 1$: $\frac{dy}{dx} = 5u^4 \cdot 3 = 15(3x + 1)^4$.

1.7 — Single-variable marginals (Together) Together

Differentiate (one variable at a time).

1. Cubic cost: $C(q) = q^3 - 6q^2 + 15q + 100$. $C'(q) = 3q^2 - 12q + 15$ MC

2. Monopolist: $p(q) = 120 - 2q$, $C(q) = q^2 + 10q + 50$. Profit $\pi(q) = q(120 - 2q) - C(q) = 110q - 3q^2 - 50$.
 $\pi'(q) = 110 - 6q$ FOC: $q^* = 110/6 \approx 18.3$

3. Cobb–Douglas production: $Y(K) = 3K^{0.6}$. $Y'(K) = 1.8 K^{-0.4}$ MP_K

4. Fixed cost + composite cost: $TC(q) = 100 + 20(2q + 1)^3$. $TC'(q) = 120(2q + 1)^2$ chain rule

1.7 Euler's theorem

Setup (from 1.6). $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is C^1 , homogeneous of degree k : $f(t\mathbf{x}) = t^k f(\mathbf{x})$ for $t > 0$.

Euler's theorem. $\sum_{i=1}^n x_i \frac{\partial f}{\partial x_i}(\mathbf{x}) = k f(\mathbf{x})$.

Proof. Fix $\mathbf{x}$, $\varphi(t) = f(t\mathbf{x}) = t^k f(\mathbf{x})$.

RHS: $\varphi'(t) = k t^{k-1} f(\mathbf{x})$. **LHS (chain rule):** $\varphi'(t) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(t\mathbf{x}) x_i$.

At $t = 1$: $\sum_i x_i f_{x_i}(\mathbf{x}) = k f(\mathbf{x})$.

Multivariate chain rule; partials defined formally in 1.8.

1.7 Euler's theorem — the adding-up problem

Economic question. In a competitive economy, if labor and capital are each paid its **marginal product**, do total factor payments exactly equal total output? Or is there surplus profit or a deficit?

This is the **adding-up problem** (Clark, Wicksteed). If payments do not exhaust output, marginal productivity theory of distribution looks inconsistent.

Mathematical condition. Production $Q = f(K, L)$, CRS (degree 1): $f(tK, tL) = t f(K, L)$ for $t > 0$. Euler's theorem ($k = 1$):

$$Q = \frac{\partial Q}{\partial K} K + \frac{\partial Q}{\partial L} L = MP_K \cdot K + MP_L \cdot L.$$

Algebra only so far — no claim yet about wages or rents.

1.7 Euler's theorem — competitive factor markets

Total factor payments: $rK + wL$.

Perfect competition in factor markets $\Rightarrow$ each factor is paid its **value marginal product**:

$$r = P \frac{\partial Q}{\partial K}, \quad w = P \frac{\partial Q}{\partial L}.$$

Euler (CRS, $k = 1$). $Q = \frac{\partial Q}{\partial K}K + \frac{\partial Q}{\partial L}L$. Multiply by P :

$$PQ = rK + wL.$$

Revenue equals factor payments — output **exactly exhausted**; zero profit.

If returns to scale $\neq 1$, see IRS/DRS next.

1.7 Euler's theorem — IRS and DRS

CRS ($k = 1$). $PQ = rK + wL$ — payments exhaust revenue; zero profit.

IRS ($k > 1$). Paying each factor its marginal product would cost **more** than revenue — positive profit wedge (or inconsistency with perfect competition).

DRS ($k < 1$). MP payments **fall short** of revenue — output not fully exhausted; surplus remains.

Intuition: under IRS, marginal contributions add up to more than current output; under DRS, to less.

1.7 Euler's theorem — why IRS implies losses

Better technology, IRS. Production has **synergy** among inputs: at a given scale, doubling all inputs more than doubles output — the extra is synergy.

Marginal product traps synergy. $MP_K = \partial Q / \partial K$ holds L (and other inputs) **fixed** at current levels. Those inputs are already in place, so an extra unit of K gets credit for synergy between the new capital and existing labor.

Repeat for every machine and every worker: the **same synergy** is credited to each input. Summing $MP_K \cdot K + MP_L \cdot L$ therefore **counts synergy several times**.

Euler. Under IRS ($k > 1$), $MP_K \cdot K + MP_L \cdot L = k Q > Q$. If each factor is paid its MP, total payments exceed revenue — the firm would run **losses** under perfect competition.

1.8 Level curves

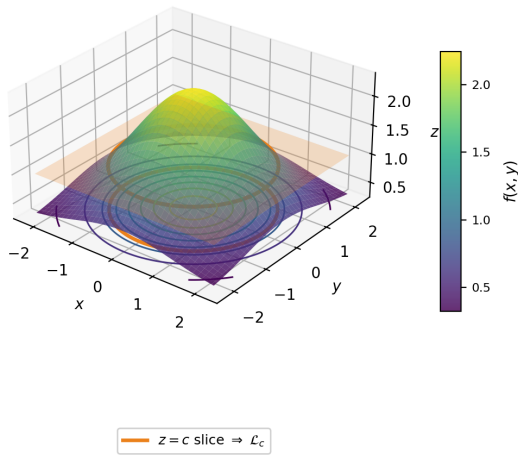
Multivariate function: $z = f(x, y)$, e.g. utility $U(x_1, x_2)$.

Level curve at height c :

$$\mathcal{L}_c = \{(x, y) \in \mathbb{R}^2 : f(x, y) = c\}.$$

On a hill/mountain graph, slice the surface by the horizontal plane $z = c$; the slice projects to a level curve in the (x, y) -plane.

Contour map: top-down picture of several $\mathcal{L}_c$'s. Along a level curve, f is **constant**.



1.8 Multivariate calculus — roadmap

Question: how does $f(x, y)$ change when inputs move?

- Hold one variable fixed $\Rightarrow$ **partial derivatives**
- Small joint change $(dx, dy) \Rightarrow$ **total differential**
- Move in a chosen direction $\mathbf{u} \Rightarrow$ **directional derivative** (rate version of df)
- Differentiate in two orders $\Rightarrow$ **cross-partials** (Schwarz)

1.8 Partial derivatives

Definition. For $f(x, y)$,

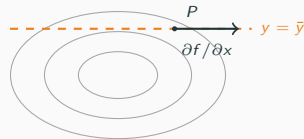
$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h},$$

holding $y = b$ fixed (and analogously $\partial f / \partial y$).

On a contour map: fix $y = \bar{y}$ (horizontal line).

The slope along that line is $\partial f / \partial x$.

Econ: $MP_K = \partial Y / \partial K$ with L fixed.



1.8 Total differential

Goal: approximate how f changes when (x, y) moves a little from (x_0, y_0) by $(\Delta x, \Delta y)$.

1. **Exact change.** $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$.

2. **Add/subtract** at $(x_0 + \Delta x, y_0)$:

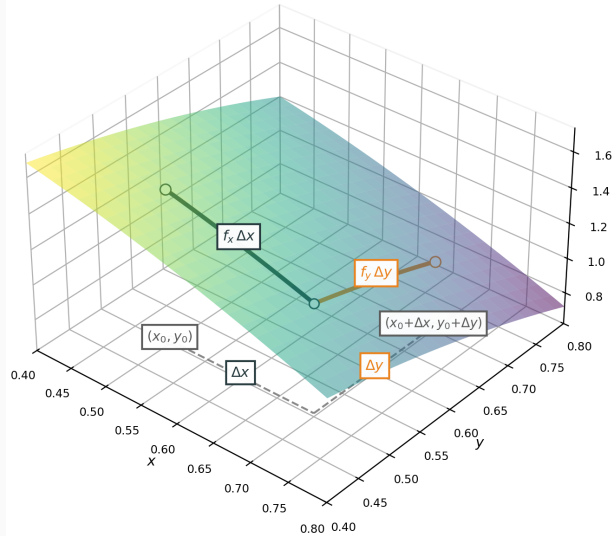
$$\Delta z = [f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0)] + [f(x_0 + \Delta x, y_0) - f(x_0, y_0)].$$

3. **Mean value theorem** ($\theta_1, \theta_2 \in (0, 1)$):

$$f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) = f_y(x_0 + \Delta x, y_0 + \theta_1 \Delta y) \Delta y,$$

$$f(x_0 + \Delta x, y_0) - f(x_0, y_0) = f_x(x_0 + \theta_2 \Delta x, y_0) \Delta x.$$

4. **Small steps** $\Delta x, \Delta y \rightarrow 0$: $\Delta z \approx f_x \Delta x + f_y \Delta y =: dz, \quad \nabla f = (f_x, f_y)$.



1.8 Using the total differential

When: approximate how output changes when **several inputs** move a little (comparative statics, linearization).

Example. Cobb–Douglas production $Y = 3K^\alpha L^{1-\alpha}$ with $\alpha = 0.5$, at $(K_0, L_0) = (100, 64)$, $dK = 2$, $dL = 1$:

$$dY = \underbrace{1.5K^{-0.5}L^{0.5}}_{MP_K} dK + \underbrace{1.5K^{0.5}L^{-0.5}}_{MP_L} dL.$$
$$dY = \frac{1.5}{10} \cdot 8 \cdot 2 + \frac{1.5 \cdot 10}{8} \cdot 1 = 2.4 + 1.875 \approx 4.28.$$

Exact: $Y(100, 64) = 240$, $Y(102, 65) = 3\sqrt{6630} \approx 244.27$, so $\Delta Y \approx 4.27$ (close to dY when dK, dL are small).

1.8 Gradient and level curves

At P on $\mathcal{L}_c$: $\nabla f(P) \perp \mathcal{L}_c$ and points uphill.

1. **Perpendicular.** Path on the curve:

$\mathbf{r}(t) = (x(t), y(t))$; tangent

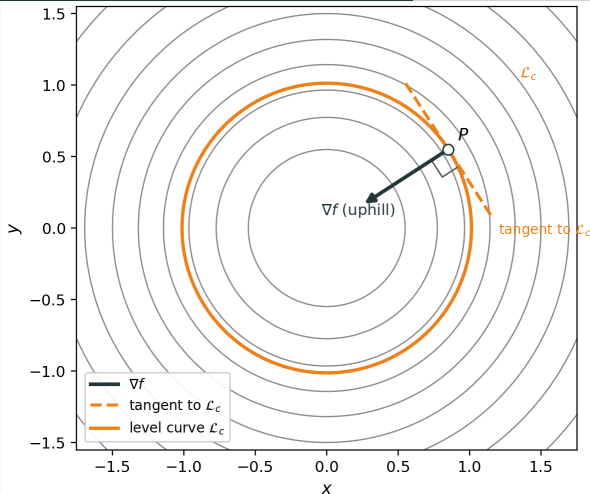
$\mathbf{r}'(t) = (x'(t), y'(t))$. On $\mathcal{L}_c$, $f(\mathbf{r}(t)) = c$,
so

$$0 = \frac{d}{dt} f(\mathbf{r}(t)) = \nabla f \cdot \mathbf{r}'(t) \Rightarrow \nabla f \perp \mathbf{r}'(t).$$

2. **Uphill.** For a small move $\mathbf{h} = (dx, dy)$,

$$\begin{aligned} df &\approx f_x dx + f_y dy = \nabla f \cdot \mathbf{h} \\ &= \sqrt{dx^2 + dy^2} \sqrt{f_x^2 + f_y^2} \cos \theta, \end{aligned}$$

with θ the angle between $\mathbf{h}$ and ∇f . ∇f is uphill: $\mathbf{h} \parallel \nabla f \Rightarrow \cos \theta = 1$, so df is largest and positive.



1.8 Directional derivative

Definition. For unit direction $\mathbf{u} = (u_1, u_2)$, $\|\mathbf{u}\| = 1$, the **directional derivative** at (a, b) is

$$D_{\mathbf{u}}f(a, b) := \lim_{t \rightarrow 0} \frac{f(a + tu_1, b + tu_2) - f(a, b)}{t}$$

(rate of change of f per unit step along $\mathbf{u}$).

Derive. Step $(dx, dy) = t\mathbf{u}$. By the total differential, for small t ,

$$f(a + tu_1, b + tu_2) - f(a, b) \approx f_x(tu_1) + f_y(tu_2) = t \nabla f \cdot \mathbf{u}.$$

Divide by t and let $t \rightarrow 0$:

$$D_{\mathbf{u}}f(a, b) = \nabla f \cdot \mathbf{u} = \|\nabla f\| \cos \theta.$$

1.8 Example: targeted input change — question

A firm's output with a targeted input change.

Cobb–Douglas production (capital K , labor L):

$$Q(K, L) = K^\alpha L^{1-\alpha}, \quad \alpha = 0.5.$$

Current inputs: $(K_0, L_0) = (100, 100)$.

A subsidy is paid only if the firm increases **both** inputs in ratio 2:1 (for every 1 unit of labor, add 2 units of capital).

This defines direction $\mathbf{v} = (2, 1)$ in (K, L) -space.

Question: How fast does output Q change per unit step in that direction?

1.8 Example: targeted input change — solution

1. Gradient at (100, 100). $Q_K = 0.5K^{-0.5}L^{0.5}$, $Q_L = 0.5K^{0.5}L^{-0.5}$. Since $100^{0.5} = 10$ and $100^{-0.5} = 0.1$,

$$Q_K = 0.5(0.1)(10) = 0.5, \quad Q_L = 0.5(10)(0.1) = 0.5.$$

So $\nabla Q = (0.5, 0.5)$.

2. Unit direction. $\|\mathbf{v}\| = \sqrt{5}$,

$$\mathbf{u} = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right).$$

3. Directional derivative.

$$D_{\mathbf{u}}Q = \nabla Q \cdot \mathbf{u} = 0.5 \left(\frac{2}{\sqrt{5}} + \frac{1}{\sqrt{5}} \right) = \frac{1.5}{\sqrt{5}} \approx 0.67.$$

Interpretation: from (100, 100), moving in direction $\mathbf{u}$ (ratio 2:1), output rises at initial rate ≈ 0.67 units of Q per unit distance in input space.

1.8 Cross-partial derivatives and Schwarz

Cross-partials. For $f(x, y)$,

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right), \quad f_{yx} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right).$$

Schwarz's theorem (stated). If f_{xy} and f_{yx} exist and are continuous near (a, b) , then $f_{xy}(a, b) = f_{yx}(a, b)$.

Example. $Y = K^\alpha L^{1-\alpha}$:

$$\frac{\partial MP_K}{\partial L} = Y_{KL} = \alpha(1-\alpha)K^{\alpha-1}L^{-\alpha} = \frac{\partial MP_L}{\partial K} = Y_{LK}.$$

Econ: Y_{KL} tells how MP_K changes when L rises (and vice versa). Sign of Y_{KL} : complements (+) vs. substitutes (-).

1.9 Limits — why this section

Berkeley's critique (1734). Early calculus treated h as “infinitely small”: nonzero when dividing, then set to 0. Berkeley called these “**ghosts of departed quantities**” — logically suspicious.

Example: $f(x) = x^2$ at $a = 3$.

$$\frac{f(3+h) - f(3)}{h} = \frac{(3+h)^2 - 9}{h} = \frac{6h + h^2}{h} = 6 + h \quad (h \neq 0).$$

Early calculus: “let $h \rightarrow 0$ ” to get $f'(3) = 6$.

Berkeley's objection. The algebra **requires** $h \neq 0$ to divide by h ; then we act as if $h = 0$ at the end. That is the ghost: h is treated as both present and absent.

1.9 Limit of a sequence (ε - N)

Let (a_n) be a sequence and $L \in \mathbb{R}$.

Definition. $\lim_{n \rightarrow \infty} a_n = L$ means:

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n \geq N : \quad |a_n - L| < \varepsilon.$$

Reading the quantifiers.

- ε : target accuracy (how close to L you want).
- N : step after which the sequence stays inside that accuracy.
- Order matters: you choose ε first, *then* find N .

If the limit exists, the sequence **converges** to L ; otherwise it **diverges**.

1.9 Limit of a function (ε - δ)

$\lim_{x \rightarrow a} f(x) = L$ means $\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D: 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

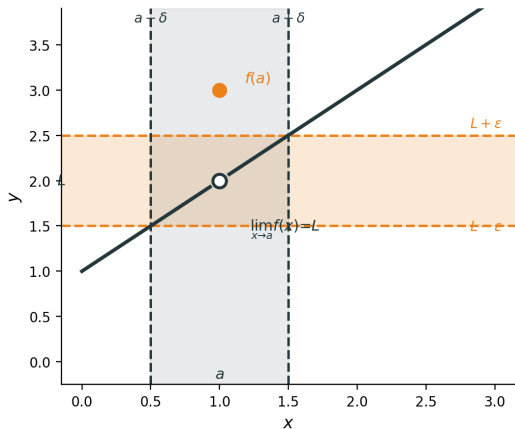
Example. $f(x) = x + 1$ for $x \neq 1$; $f(1) = 3$.

$$\lim_{x \rightarrow 1} f(x) = 2, \quad f(1) = 3 \neq 2.$$

For $\varepsilon = \frac{1}{2}$, take $\delta = \frac{1}{2}$.

If $0 < |x - 1| < \delta$, then

$$|f(x) - 2| = |x - 1| < \varepsilon.$$



On $0 < |x - a| < \delta$, the curve stays in the ε -band around L ; the value at $x = a$ is ignored.

1.9 Continuous and differentiable

Continuous at a . f is continuous at a if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

So the limit exists **and** equals the function value at a .

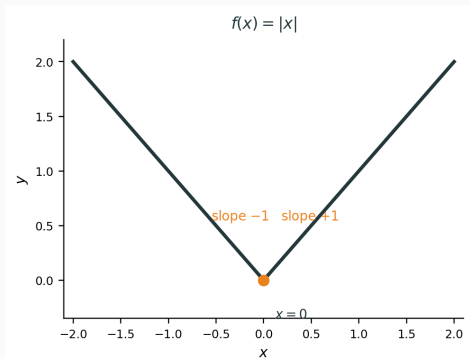
Differentiable at a . f is differentiable at a if

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

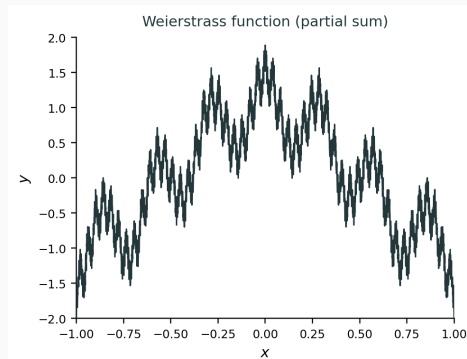
exists (finite).

Relation. Differentiable at $a \Rightarrow$ continuous at a . Converse fails: see examples on the next slide.

1.9 Continuous but not differentiable — for interest



$f(x) = |x|$. Continuous on $\mathbb{R}$; at 0, left slope -1 and right slope $+1$ disagree $\Rightarrow$ not differentiable at 0.



Weierstrass (1872). Some $W: \mathbb{R} \rightarrow \mathbb{R}$ is **continuous everywhere** but **differentiable nowhere**. Plot: 10 terms of $\sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$, $a = \frac{1}{2}$, $b = 7$.

Practice.

1.9 — Limits practice **Together**

Practice.

1. Let $f(x) = x^2$. Use ε - δ to prove $f'(a) = 2a$.

1.9 — Limits practice Together

Practice.

1. Let $f(x) = x^2$. Use ε - δ to prove $f'(a) = 2a$.

Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |h| < \delta$, then $\left| \frac{f(a+h) - f(a)}{h} - 2a \right| < \varepsilon$.

1.9 — Limits practice Together

Practice.

1. Let $f(x) = x^2$. Use ε - δ to prove $f'(a) = 2a$.

Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |h| < \delta$, then $\left| \frac{f(a+h) - f(a)}{h} - 2a \right| < \varepsilon$.

Algebra. For $h \neq 0$, $\frac{(a+h)^2 - a^2}{h} = \frac{2ah + h^2}{h} = 2a + h$.

1.9 — Limits practice Together

Practice.

1. Let $f(x) = x^2$. Use ε - δ to prove $f'(a) = 2a$.

Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |h| < \delta$, then $\left| \frac{f(a+h)-f(a)}{h} - 2a \right| < \varepsilon$.

Algebra. For $h \neq 0$, $\frac{(a+h)^2 - a^2}{h} = \frac{2ah + h^2}{h} = 2a + h$.

Choose $\delta = \varepsilon$. Then $\left| \frac{f(a+h)-f(a)}{h} - 2a \right| = |h| < \varepsilon$.

so $f'(a) = 2a$

1.9 — Limits practice Together

Practice.

1. Let $f(x) = x^2$. Use ε - δ to prove $f'(a) = 2a$.

Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |h| < \delta$, then $\left| \frac{f(a+h)-f(a)}{h} - 2a \right| < \varepsilon$.

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so $f'(a) = 2a$

2. $f(x) = x + 1$ for $x \neq 1$; $f(1) = 3$. Use ε - δ to prove $\lim_{x \rightarrow 1} f(x) = 2$.

1.9 — Limits practice Together

Practice.

1. Let $f(x) = x^2$. Use ε - δ to prove $f'(a) = 2a$.

Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |h| < \delta$, then $\left| \frac{f(a+h)-f(a)}{h} - 2a \right| < \varepsilon$.

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so $f'(a) = 2a$

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Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |x - 1| < \delta$, then $|f(x) - 2| < \varepsilon$.

1.9 — Limits practice Together

Practice.

1. Let $f(x) = x^2$. Use ε - δ to prove $f'(a) = 2a$.

Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |h| < \delta$, then $\left| \frac{f(a+h)-f(a)}{h} - 2a \right| < \varepsilon$.

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Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |x - 1| < \delta$, then $|f(x) - 2| < \varepsilon$.

Algebra. For $x \neq 1$ near 1, $f(x) = x + 1$, so $|f(x) - 2| = |x - 1|$.

1.9 — Limits practice Together

Practice.

1. Let $f(x) = x^2$. Use ε - δ to prove $f'(a) = 2a$.

Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |h| < \delta$, then $\left| \frac{f(a+h)-f(a)}{h} - 2a \right| < \varepsilon$.

Algebra. For $h \neq 0$, $\frac{(a+h)^2 - a^2}{h} = \frac{2ah + h^2}{h} = 2a + h$.

Choose $\delta = \varepsilon$. Then $\left| \frac{f(a+h)-f(a)}{h} - 2a \right| = |h| < \varepsilon$.

so $f'(a) = 2a$

2. $f(x) = x + 1$ for $x \neq 1$; $f(1) = 3$. Use ε - δ to prove $\lim_{x \rightarrow 1} f(x) = 2$.

Goal. $\forall \varepsilon > 0 \exists \delta > 0$: if $0 < |x - 1| < \delta$, then $|f(x) - 2| < \varepsilon$.

Algebra. For $x \neq 1$ near 1, $f(x) = x + 1$, so $|f(x) - 2| = |x - 1|$.

Choose $\delta = \varepsilon$. Then $|f(x) - 2| = |x - 1| < \varepsilon$.

$f(1) = 3 \neq 2$: limit exists, but f not continuous at 1

A.1 Deriving rules (I): constants, sums, powers

Write $D_a f = f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$.

Constant: $f(x) = c \Rightarrow D_a f = \lim_{h \rightarrow 0} \frac{c - c}{h} = 0$.

Sum: $D_a(f + g) = D_a f + D_a g$ (split the numerator; limits add).

Power rule (integer $n \geq 1$): for $f(x) = x^n$,

$$\frac{(a+h)^n - a^n}{h} = \frac{a^n + na^{n-1}h + \cdots + h^n - a^n}{h} = na^{n-1} + \text{terms with extra factors of } h \xrightarrow[h \rightarrow 0]{} na^{n-1}.$$

So $(x^n)' = nx^{n-1}$.

A.2 Deriving rules (II): product and quotient

Product rule. For $h(x) = f(x)g(x)$,

$$\frac{h(a+h) - h(a)}{h} = f(a+h) \frac{g(a+h) - g(a)}{h} + g(a) \frac{f(a+h) - f(a)}{h} \xrightarrow{h \rightarrow 0} f(a)g'(a) + f'(a)g(a).$$

So $(fg)' = f'g + fg'$.

Quotient rule. Write $\frac{f}{g} = f \cdot g^{-1}$. For $y = 1/g$, $\ln y = -\ln g$, so $\frac{y'}{y} = -\frac{g'}{g} \Rightarrow y' = -\frac{g'}{g^2}$.

Then product rule gives $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$.

A.3 Key facts for exp and log

Fact 1: $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$

Why? Start with $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ and $e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$. Binomial expansion:

$$\left(1 + \frac{h}{n}\right)^n = 1 + h + \frac{n(n-1)}{2n^2}h^2 + \dots \xrightarrow{n \rightarrow \infty} e^h.$$

So $e^h = 1 + h + O(h^2)$, hence $\frac{e^h - 1}{h} = 1 + O(h) \rightarrow 1.$

Fact 2: $\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1.$

Why? For small u , Fact 1 gives $e^u \approx 1 + u$. If $v = \ln(1+u)$, then $e^v = 1 + u$, so $v \approx u$ and $\frac{\ln(1+u)}{u} \approx 1$. Equivalently, Taylor: $\ln(1+u) = u - \frac{u^2}{2} + \dots$.

A.4 Deriving the exponential: $(e^x)' = e^x$

Use Fact 1 from the previous slide: $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$.

For $f(x) = e^x$,

$$f'(a) = \lim_{h \rightarrow 0} \frac{e^{a+h} - e^a}{h} = e^a \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = e^a.$$

So $(e^x)' = e^x$.

Reading: a small step h multiplies e^x by about $1 + h$.

A.5 Deriving the logarithm: $(\ln x)' = \frac{1}{x}$

Method 1 (inverse function). If $y = \ln x$, then $e^y = x$. Differentiate: $e^y y' = 1$, so $y' = \frac{1}{e^y} = \frac{1}{x}$.

Method 2 (limit definition). For $a > 0$, set $u = h/a$:

$$\frac{\ln(a+h) - \ln a}{h} = \frac{1}{h} \ln\left(1 + \frac{h}{a}\right) = \frac{1}{a} \cdot \frac{\ln(1+u)}{u} \xrightarrow{u \rightarrow 0} \frac{1}{a},$$

by Fact 2. So $(\ln x)' = \frac{1}{x}$.